EXACT AND APPROXIMATE KOBAYASHI CHAINS ON STRONGLY PSEUDOCONVEX DOMAINS

Abstract. We distinguish finite realization of the Kobayashi distance from uniform approximation by finite chains. We construct a bounded, star-shaped domain in $\mathbb{C}^2$ with C^∞ strongly pseudoconvex boundary and a compact family of endpoint pairs whose minimum numbers of discs in exactly minimizing chains are unbounded. A limiting pair admits no finite realizing chain. This gives a negative answer to Krantz's finite-chain question. Nevertheless, the same entire obstruction family has double-exponentially small upper bounds for its finite-chain errors and admits uniform approximation by $O(\log\log(1/\varepsilon))$ discs. This improves the elementary logarithmic bound and shows that it is not sharp; no matching lower bound for the improved rate is asserted. On the other hand, for every bounded strongly pseudoconvex domain $\Omega \Subset \mathbb{C}^n$ with C^2 boundary, every pair of points can be joined by a chain of at most $2 + (2n+3)\lceil A_\Omega/\varepsilon\rceil$ discs with cost less than $k_\Omega + \varepsilon$, uniformly for $0 < \varepsilon \le 1$. The proof combines a stationary-disc identity for moving centers, finite-dimensional moment matching, and a weighted subdivision of real Kobayashi geodesics.

1. Introduction and main results

Let Δ be the unit disc and normalize its Poincaré distance by

$$\rho_\Delta(\alpha,\beta) = \operatorname{arctanh}\left|\frac{\alpha-\beta}{1-\overline{\alpha}\beta}\right|.$$

A Kobayashi chain from z to w in a domain Ω consists of maps $f_j \in \mathcal{O}(\Delta,\Omega)$ and marked points $\alpha_j, \beta_j \in \Delta$ with

$$f_1(\alpha_1) = z, \quad f_j(\beta_j) = f_{j+1}(\alpha_{j+1}), \quad f_m(\beta_m) = w.$$

Its cost is $\sum_{j=1}^m \rho_\Delta(\alpha_j,\beta_j)$. We write $k_\Omega^{(m)}$ for the infimum over chains with at most m discs, $\ell_\Omega = k_\Omega^{(1)}$, and $k_\Omega = \inf_m k_\Omega^{(m)}$. Constant links may be inserted. The chain distance agrees with the integrated Kobayashi–Royden metric; see [18, 13, 14]. We use "number of discs" for combinatorial length and "cost" for metric length. Our infinitesimal conventions are

$$\kappa_\Omega(z;v) = \inf\{t>0 : \exists f \in \mathcal{O}(\Delta,\Omega),\ f(0)=z,\ tf'(0)=v\},$$
$$\gamma_\Omega(z;v) = \sup\{|dF_z(v)| : F \in \mathcal{O}(\Omega,\Delta),\ F(z)=0\}.$$

Date: October 23, 2026.

2020 *Mathematics Subject Classification.* Primary 32F45; Secondary 32T15, 32H02.

Key words and phrases. Kobayashi distance, Lempert function, strongly pseudoconvex domain, stationary disc, finite chains.

Krantz asked whether a strongly pseudoconvex domain admits a fixed finite bound for exact realization of all its Kobayashi distances. The question is recorded by Pang [16, p. 124] and as Prize Problem 1 in [3, §14]. Nikolov and Pflug [13, Remark (iii) after Corollary 2] formulate it as $k_D = k_D^{(m(D))}$. Our first result rules out even pointwise finite realization in this class.

Theorem 1.1 (Failure of finite realization)**.** *There exist a bounded, star-shaped domain $\Omega \subset \mathbb{C}^2$ with C^∞ strongly pseudoconvex boundary, a compact set $C \Subset \Omega$, and points $p, q_*, q_N \in C$, $N \geq 1$, with $q_N \to q_*$, such that:*

1. *$k_\Omega(p, q_N)$ is realized by an N-disc chain and by no chain with fewer discs;*
2. *no finite chain realizes $k_\Omega(p, q_*)$;*
3. *$k_\Omega^{(m)}(p, q_*) > k_\Omega(p, q_*)$ for every finite m.*

The exact obstruction is compatible with rapid approximation on the same family.

Corollary 1.2 (Exponential approximation on the obstruction family)**.** *For the domain and endpoints constructed in Theorem 1.1, put*

$$\mathcal{Q} = \{q_*\} \cup \{q_N : N \geq 1\}, \qquad \eta = 10^{-12}, \quad \sigma = \frac{1}{256}, \quad C_{\log} = 32\eta\sigma.$$

For every integer $m \geq 1$,

$$0 \leq \sup_{q \in \mathcal{Q}} (k_\Omega^{(m+1)}(p, q) - k_\Omega(p, q)) \leq C_{\log} 4^{-m}. \tag{1.1}$$

For every $\varepsilon > 0$ and every $q \in \mathcal{Q}$, an actual chain from p to q has cost strictly less than $k_\Omega(p, q) + \varepsilon$ and uses at most

$$2 + \max\left\{0, \left\lceil \log_4 \frac{C_{\log}}{\varepsilon} \right\rceil\right\} \tag{1.2}$$

discs.

Thus the logarithmic upper bound is uniform on $\{p\} \times \mathcal{Q}$ (and on its transpose). It is not asserted on all of $\Omega \times \Omega$.

The flat function used to select the graph sections permits a stronger estimate.

Theorem 1.3 (Double-exponential errors on the obstruction family)**.** *For the same domain and family $\mathcal{Q}$, set*

$$C_{\text{flat}} = 64\eta^2\sigma, \qquad c_{\text{flat}} = \frac{1}{36\sigma^2}.$$

There is an integer $m_0 \geq 2$ such that, for every $m \geq m_0$ and every $q \in \mathcal{Q}$, an actual chain with at most $m + 1$ discs has cost at most

$$k_\Omega(p, q) + C_{\text{flat}} \exp(-c_{\text{flat}} 4^m).$$

In particular,

$$\sup_{q \in \mathcal{Q}} (k_\Omega^{(m+1)}(p, q) - k_\Omega(p, q)) \leq C_{\text{flat}} \exp(-c_{\text{flat}} 4^m) \qquad (m \geq m_0). \tag{1.3}$$

For every $\varepsilon > 0$, strict additive approximation within ε is therefore possible, uniformly on $\{p\} \times \mathcal{Q}$, with at most

$$1 + \max\left\{m_0,\ \left\lceil \log_4 \left[c_{\mathrm{flat}}^{-1} \log\left(1 + \frac{2C_{\mathrm{flat}}}{\varepsilon}\right)\right]\right\rceil\right\} \tag{1.4}$$

discs. Thus the count is $O(\log\log(1/\varepsilon))$ as $\varepsilon \downarrow 0$, and the logarithmic bound of Corollary 1.2 is not sharp.

Here m_0 absorbs a uniform implicit-function neighborhood and a fixed size restriction on the auxiliary discs. No numerical value for this threshold is asserted. The theorem concerns the specified obstruction family; neither a whole-domain double-logarithmic bound nor sharpness of this improved rate is claimed.

The general positive result quantifies approximation uniformly over the entire domain.

Theorem 1.4 (Uniform quantitative approximation)**.** *Let $\Omega \Subset \mathbb{C}^n$ be a bounded domain with C^2 strongly pseudoconvex boundary. There exists $A_\Omega > 0$ such that, for every $0 < \varepsilon \le 1$ and every $z, w \in \Omega$, an actual Kobayashi chain from z to w has cost strictly less than $k_\Omega(z,w) + \varepsilon$ and uses at most*

$$2 + (2n+3)\left\lceil \frac{A_\Omega}{\varepsilon} \right\rceil \tag{1.5}$$

discs. For $n = 1$ one may take $A_\Omega = 1$; for $n \ge 2$ it is specified in (3.45) in terms of fixed geometric and stationary-disc constants.

Consequently, $k_\Omega^{(m)} \to k_\Omega$ uniformly on $\Omega \times \Omega$, with additive error $O_\Omega(m^{-1})$. For instance, take $\varepsilon = 2(2n+3)A_\Omega/m$ for all sufficiently large m. The constant A_Ω is explicit relative to the inputs fixed below; no formula solely in a defining function's C^2 norm and a Levi lower bound is asserted. Nor do we claim that the exponent 1 is optimal.

1.1. **Relation to earlier work.** Lempert's theorem gives $k_\Omega = \ell_\Omega$ on convex domains [10, 11]. Pang constructed strongly pseudoconvex domains where local one-disc realization fails [16, Corollary 4.13 and Example 4.22]; this does not exclude realization by a longer finite chain. The boundary model underlying our construction is already present in Pang [15, Example 3.28, equation (3.29), pp. 973–974]: taking $k = 2$ and $\lambda = 1$ there gives exactly the function ρ_0 in (2.7). Pang's Example 3.2, pp. 968–969, credited there to Sibony, also gives nonisolated infinitesimally extremal discs with the same center and tangent vector, using the related model with $\bar{z}^4 w^2$. See also [2]. Neither the starting defining function nor the projection mechanism is claimed as new. Our additional construction prescribes a countable family of affine sections and an intersection pattern forcing arbitrarily many exact transitions, while retaining boundedness and smooth strong pseudoconvexity.

Corollary 1.2 follows from the geometrically decreasing endpoint heights and one vertical correction disc. The stronger Theorem 1.3 uses auxiliary

holomorphic graphs satisfying the continuous boundary relation, followed by a radial contraction controlled by the flat boundary term. Both estimates concern the specified obstruction family.

Pang's local deformation theory is another antecedent. His Theorem 3.23, p. 971, gives isolated local extremality under coercivity of the second variation; Theorem 4.8, pp. 977–978, gives smooth deformation under nondegeneracy. Theorems 5.18–5.19 and Corollary 5.20, pp. 985–986, give stability of circular representations and smoothness of the infinitesimal Kobayashi metric near the base point, including strongly pseudoconvex complete circular domains [15]. These concern individual infinitesimally extremal discs and local metric regularity, and do not assert finite-chain realization or error-dependent chain counts. See [17, pp. 1294–1295] for later developments and historical discussion.

Kosiński [6, Theorems 1–3, pp. 273, 276] establishes equality of invariant functions on selected nonempty open subsets and existence of boundary-attached complex geodesics. These selected-pair results do not assert an all-pair bound on finite chains.

Nikolov–Pflug [13, Theorem 1 and Corollary 2] show that $2n-1$ summands suffice to convexify the infinitesimal Kobayashi indicatrix, and that $k_D^{(2n-1)}(z,w)/k_D(z,w) \to 1$ locally uniformly near the diagonal on taut domains. These first-order statements neither imply finite exact realization nor supply the additive, global rate of Theorem 1.4. Attainment and continuity for each fixed finite chain number on taut domains are established in [14, Proposition 4(b)]; we include a direct attainment argument for our example to make the strict inequality in Theorem 1.1 transparent.

Krantz [9, Proposition 17, pp. 11–12, and Theorem 18, pp. 12–13] proposes a lossless disc-merging argument and states a uniform approximation assertion for one positive tolerance. The published theorem explicitly bounds the chain by $K = K(\Omega)$, correcting a notation omission in the earlier preprint [8]; it does not quantify arbitrary tolerances or give an error-dependent rate. Corollary 2.4 below excludes the uniform lossless interpolation rule used in that argument, for a precise fixed normalization of the marked parameters. It does not refute the literal fixed-positive-error conclusion of Theorem 18, which follows from Theorem 1.4 here. No assertion against the Docquier–Grauert retraction theorem is involved.

The approximation proof uses established boundary geometry from Balogh–Bonk [1], quadratic localization from Nikolov [12], and stationary-disc theory at C^2 regularity from Kosiński–Warszawski [7]. The additional local estimate proved here is

$$h = k_\Omega(z,w) \le c_\Omega \delta(z) \quad \Longrightarrow \quad k_\Omega^{(2n+3)}(z,w) \le h + C_\Omega \sqrt{\delta(z)}\, h^2.$$

The factor $\sqrt{\delta(z)}$ is what permits the global linear rate in ε^{-1}.

Remark 1.5 (Scope of the literature comparison). To our knowledge, the specific failure of finite realization in Theorem 1.1, the global additive rate

in Theorem 1.4, and the obstruction-family estimate in Theorem 1.3 are not contained in the sources compared above. The comparison includes the complete published texts of [15, 9]. This is a qualified priority assessment, not an exhaustive exclusion of all antecedents. In particular, Pang's starting boundary model and individual-disc theory, and Krantz's fixed-positive-error approximation assertion, are prior work. We claim neither the novelty of qualitative approximation nor optimality of the global or double-logarithmic rates. The latter estimate is uniform only on the specified family.

2. A domain with unbounded exact chain length

We prove Theorem 1.1 by prescribing all sections of a coordinate projection to Δ. Equality in Schwarz–Pick will then turn exact chain realization into a restriction on successive intersections of these sections.

Fix

$$r = \frac{1}{64}, \qquad \sigma = \frac{r}{4} = \frac{1}{256}, \qquad \eta = 10^{-12},$$

and put

$$U = \{(z, w) \in \mathbb{C}^2 : |z| < 2,\ |w| < r\}.$$

For real s, define

$$A(s) = s + i\eta s^2, \qquad B(s) = \eta s^2 + \frac{9i}{7}\eta^2 s^3, \qquad h_s(z) = A(s)z + B(s).$$

The prescribed graph discs will be

$$G_*(z) = (z, 0), \qquad G_j(z) = (z, h_{s_j}(z)), \qquad s_j = \sigma 2^{1-j}, \quad j \geq 1. \tag{2.1}$$

We construct Ω so that these are *exactly* the holomorphic sections of the coordinate projection

$$\pi : \Omega \longrightarrow \Delta, \qquad \pi(z, w) = z.$$

2.1. Cutoff functions and a discrete zero set.

We first specify a smooth cutoff. Let

$$\beta(t) = \begin{cases} \exp\left(-\dfrac{1}{t(1-t)}\right), & 0 < t < 1, \\ 0, & \text{otherwise}, \end{cases}$$

and set

$$I = \int_0^1 \beta(t)\, dt, \qquad J(t) = \frac{1}{I}\int_{-\infty}^{t} \beta(v)\, dv, \qquad \chi(t) = J(4t - 2).$$

Then χ is C^∞, takes values in $[0, 1]$, and satisfies

$$\chi(t) = 0 \quad (t \leq \tfrac{1}{2}), \qquad \chi(t) = 1 \quad (t \geq \tfrac{3}{4}). \tag{2.2}$$

We will use the numerical bounds

$$|\chi'| \leq 20, \qquad |\chi''| \leq 1300. \tag{2.3}$$

Indeed,

$$I \geq \frac{1}{3}e^{-9/2}, \qquad \sup \beta \leq e^{-4}, \qquad \sup |\beta'| \leq 16e^{-4}.$$

The last inequality follows by writing $v = 1/[t(1-t)] \geq 4$ and using

$$|\beta'(t)| \leq v^2 e^{-v} \leq 16e^{-4}.$$

These estimates give

$$|\chi'| \leq 12e^{1/2} < 20, \qquad |\chi''| \leq 768e^{1/2} < 1300.$$

Next define the nonnegative function

$$Q(u) = \begin{cases} \eta e^{-1/u^2} \sin^2(\pi \log_2(u/\sigma)), & u > 0, \\ 0, & u = 0, \\ \eta e^{-1/u^2}, & u < 0. \end{cases} \tag{2.4}$$

It is C^∞, and every derivative at zero vanishes. To see this, differentiate separately on the two half-lines. Each derivative is a finite sum of terms bounded near zero by

$$C|u|^{-M} e^{-1/u^2},$$

which tends to zero. Its zero set is exactly

$$Q^{-1}(0) = \{0\} \cup \{\sigma 2^k : k \in \mathbb{Z}\}. \tag{2.5}$$

For later estimates, on $|u| \leq 2r = 1/32$,

$$|Q(u)|,\ |Q'(u)|,\ |Q''(u)| \leq \eta. \tag{2.6}$$

Here is a quantitative verification. Write $q = Q/\eta$. On the positive half-line, let

$$v = u^{-2} \geq 1024, \qquad c = \frac{\pi}{\log 2} < 5.$$

Direct differentiation gives

$$|q'(u)| \leq (2v^{3/2} + 5v^{1/2})e^{-v} < 1,$$

and

$$|q''(u)| \leq (4v^3 + 26v^2 + 55v)e^{-v} < 1.$$

The inequalities follow, for example, from

$$e^v \geq v^2/2, \qquad e^v \geq v^4/24,$$

respectively. The negative half-line is easier, since the sine factor is absent.

2.2. **The defining function and its Levi form.** We now modify the defining function from Pang's Example 3.28, equation (3.29), with $k = 2$ and $\lambda = 1$ [15, pp. 973–974]:

$$\rho_0(z, w) = |z|^2 + |w|^2 - \mathrm{Re}(\bar{z}^{\,2} w^2) - 1. \tag{2.7}$$

Its Levi matrix is

$$H_{\rho_0} = \begin{pmatrix} 1 & -2z\bar{w} \\ -2\bar{z}w & 1 \end{pmatrix},$$

whose eigenvalues are $1 \pm 2|zw|$. Consequently,

$$H_{\rho_0} \geq (1 - 4r) I_2 = \frac{15}{16} I_2 \qquad \text{on } U. \tag{2.8}$$

Moreover, for $|\zeta| = 1$,

$$\rho_0(\zeta, w) = 2\big(\mathrm{Im}(w/\zeta)\big)^2. \tag{2.9}$$

For $|\zeta| = 1$, $|w| \le r$, write

$$\zeta = e^{i\theta}, \qquad w/\zeta = x + iy.$$

Define $u = u(\zeta, w) \in (-2r, 2r)$ as the unique solution of

$$x = F(\theta, u) := u + \eta u^2 \cos\theta + \frac{9}{7}\eta^2 u^3 \sin\theta. \tag{2.10}$$

This solution exists and depends smoothly on (ζ, w). In fact, on $|u| \le 2r$,

$$F_u \ge \frac{1}{2},$$

and

$$|F(\theta, u) - u| \le 2\eta u^2.$$

Thus the values at $u = -2r$ and $u = 2r$ strictly bracket the interval $[-r, r]$. Strict monotonicity gives uniqueness, and the implicit function theorem gives smoothness. Uniqueness also makes the solution periodic in θ, so it is globally defined on the circle.

Put

$$Y(\zeta, w) = \mathrm{Im}(w/\zeta - A(u) - B(u)/\zeta). \tag{2.11}$$

The defining equation for u matches the real parts. Therefore

$$Y(\zeta, w) = 0 \quad \Longleftrightarrow \quad w = A(u)\zeta + B(u) = h_u(\zeta). \tag{2.12}$$

Define the correction on the cylinder by

$$C(\zeta, w) = 2Y(\zeta, w)^2 + Q(u(\zeta, w)) - 2\big(\mathrm{Im}(w/\zeta)\big)^2, \tag{2.13}$$

and extend it into U using

$$\delta(z, w) = \begin{cases} \chi(|z|)\, C(z/|z|, w), & z \ne 0, \\ 0, & z = 0. \end{cases} \qquad \rho = \rho_0 + \delta. \tag{2.14}$$

Because χ vanishes near zero, δ is C^∞ throughout U. Equations (2.9), (2.13), and (2.14) give the exact identity

$$\rho(\zeta, w) = 2Y(\zeta, w)^2 + Q(u(\zeta, w)) \quad \text{when } |\zeta| = 1. \tag{2.15}$$

We verify quantitatively that this modification preserves strict plurisubharmonicity.

In the coordinates (θ, x, y), the inverse in (2.10) satisfies the following entrywise derivative bounds:

$$|Du| \le 2, \qquad |D^2 u| \le 24\eta. \tag{2.16}$$

For completeness, the bounds follow by implicit differentiation from

$$|F_\theta|,\ |F_{u\theta}|,\ |F_{\theta\theta}| \le \eta, \qquad |F_{uu}| \le 3\eta, \qquad F_u \ge \tfrac{1}{2}.$$

For example,

$$u_x = \frac{1}{F_u}, \qquad u_\theta = -\frac{F_\theta}{F_u}, \qquad u_{xx} = -\frac{F_{uu}}{F_u^3},$$

and

$$u_{\theta\theta} = -\frac{F_{\theta\theta} + 2F_{u\theta}u_\theta + F_{uu}u_\theta^2}{F_u}.$$

Writing $d = Y - y$, equation (2.11) becomes

$$d = \eta u^2(\sin\theta - 1) - \frac{9}{7}\eta^2 u^3 \cos\theta.$$

Direct differentiation, using (2.16), yields

$$|d| \le \eta, \qquad |Dd| \le 3\eta, \qquad |D^2 d| \le 60\eta. \tag{2.17}$$

Since

$$C = 4yd + 2d^2 + Q(u),$$

equations (2.6), (2.16), and (2.17) imply that every coordinate derivative of C of order at most two is bounded by 600η. For instance, the second derivatives are bounded by

$$(240r + 28)\eta + 300\eta^2 < 600\eta. \tag{2.18}$$

On $|z| \ge 1/2$, use local coordinates

$$V(z, w) = \left(\arg z,\, \mathrm{Re}(we^{-i\arg z}),\, \mathrm{Im}(we^{-i\arg z})\right).$$

Every first real coordinate derivative of V is bounded by 2, and every second derivative by 12. Thus, for $L = C \circ V$, the chain rule gives

$$|L| \le 600\eta, \qquad |DL| \le 3600\eta, \qquad |D^2 L| \le 43200\eta.$$

Also, (2.3) gives the entrywise bounds

$$|D\chi(|z|)| \le 20, \qquad |D^2\chi(|z|)| \le 1340.$$

The product rule in (2.14) therefore gives

$$|\delta| \le 600\eta, \qquad |D\delta| \le 15600\eta, \qquad |D^2\delta| < 10^6\eta. \tag{2.19}$$

These are bounds in the four real coordinates. In particular,

$$\|\nabla\delta\| \le 31200\eta, \qquad \|H_{\mathbb{R}}\delta\|_{\mathrm{op}} \le 4 \cdot 10^6\eta,$$

and hence

$$\|H_\delta\|_{\mathrm{op}} \le 2 \cdot 10^6\eta. \tag{2.20}$$

The last implication follows from the expression of the Levi form as one quarter of the sum of the real Hessian in a direction and its complex rotation.

Combining (2.8) and (2.20), with $\eta = 10^{-12}$, gives

$$H_\rho \ge \left(\frac{15}{16} - 2 \cdot 10^{-6}\right) I_2 > \frac{9}{10} I_2 \quad \text{on } U. \tag{2.21}$$

2.3. **Closing the domain and boundary regularity.** We next close the domain in the w-direction without introducing corners. Define

$$b(z,w) = \frac{1}{10}|z|^2 + \frac{16}{r^2}|w|^2 - 4. \tag{2.22}$$

To specify a regularized maximum, use the even probability density

$$\nu(v) = \frac{4}{I}\,\beta(4v + \tfrac{1}{2}),$$

supported in $[-1/8, 1/8]$, and set

$$a(t) = \int_{\mathbb{R}} |t - v|\nu(v)\,dv, \qquad \mathcal{M}(s,t) = \frac{s + t + a(s-t)}{2}. \tag{2.23}$$

The function a is C^∞, convex, and satisfies

$$a(t) \geq |t|, \qquad |a'(t)| \leq 1, \qquad a(t) = |t| \quad (|t| \geq \tfrac{1}{8}).$$

Here $a'' = 2\nu$, and the inequality $a \geq |\cdot|$ follows from convexity and the vanishing mean of ν. Consequently,

$$\mathcal{M}(s,t) \geq \max(s,t),$$

$$\mathcal{M}(s,t) = \max(s,t) \quad \text{if } |s-t| \geq \tfrac{1}{8}, \tag{2.24}$$

and

$$\mathcal{M}_s, \mathcal{M}_t \geq 0, \qquad \mathcal{M}_s + \mathcal{M}_t = 1. \tag{2.25}$$

Set

$$R = \mathcal{M}(\rho, b), \qquad \Omega = \{(z,w) \in U : R(z,w) < 0\}. \tag{2.26}$$

The Levi chain rule gives, for a complex tangent vector v,

$$\mathcal{L}_R(v) = \mathcal{M}_s\,\mathcal{L}_\rho(v) + \mathcal{M}_t\,\mathcal{L}_b(v) + \frac{a''(\rho - b)}{2}\,|\partial\rho(v) - \partial b(v)|^2\,.$$

Since $H_b \geq \frac{1}{10} I_2$, equations (2.21) and (2.25) imply

$$H_R \geq \frac{1}{10} I_2 \quad \text{throughout } U. \tag{2.27}$$

We check boundedness, boundary regularity, and connectedness explicitly.

For $1 \leq t < 2$, $|\zeta| = 1$, $|w| < r$, the cutoff equals one both at $t\zeta$ and at ζ. Thus

$$\rho(t\zeta, w) = \rho(\zeta, w) + (t^2 - 1)\big[1 - \mathrm{Re}(\bar{\zeta}^2 w^2)\big]. \tag{2.28}$$

The first term is nonnegative by (2.15), and the bracket is positive. Therefore $\rho < 0$ implies $|z| < 1$. Since $R \geq \rho, b$, it follows that

$$\Omega \subset \{|z| < 1,\ |w| < r/2\}, \qquad \overline{\Omega} \Subset U. \tag{2.29}$$

In particular, no part of the boundary comes from the auxiliary set U.

Let $\mathcal{E}$ denote the real radial vector field. From (2.7),

$$\mathcal{E}\rho_0 = 2(|z|^2 + |w|^2) - 4\,\mathrm{Re}(\bar{z}^2 w^2) \geq 2|z|^2(1 - 2r^2) + 2|w|^2. \tag{2.30}$$

Where the correction is supported, $|z| \geq 1/2$, so the right side exceeds 0.4997. On the other hand, (2.19) gives

$$|\mathcal{E}\delta| \leq \sqrt{4+r^2}\,\|\nabla\delta\| < 10^5\eta = 10^{-7}.$$

Thus $\mathcal{E}\rho > 0$ away from the origin. Where the correction vanishes, this follows directly from (2.30). Also,

$$\mathcal{E}b = \frac{1}{5}|z|^2 + \frac{32}{r^2}|w|^2 > 0$$

away from zero. Hence (2.25) implies

$$\mathcal{E}R > 0 \qquad \text{on } U \setminus \{0\}. \tag{2.31}$$

At the origin,

$$R(0) = \mathcal{M}(-1,-4) = -1.$$

It follows that $dR \neq 0$ on $\{R = 0\}$. Moreover, R increases strictly along every radial ray, so Ω is star-shaped and connected. Equations (2.27), (2.29), and (2.31) prove that Ω is bounded and has C^∞ strongly pseudoconvex boundary.

We will use the concrete compact subset

$$C = \overline{B}_{\mathbb{C}^2}(0,\sigma). \tag{2.32}$$

It lies in Ω. Indeed, the correction vanishes on this ball,

$$-1 \leq \rho_0 \leq \sigma^2 + \sigma^4 - 1 < 0,$$

and

$$b \leq \frac{16}{r^2}\sigma^2 - 4 = -3.$$

Thus $\rho_0 - b \geq 2$, so (2.24) gives $R = \rho_0 < 0$ throughout C.

2.4. **Classification of all sections.** We now establish the promised exact classification of sections.

All the discs in (2.1) lie in Ω. For $0 < s \leq \sigma$,

$$|A(s)| + |B(s)| \leq s + 2\eta s^2 + \frac{9}{7}\eta^2 s^3 < 1.01s. \tag{2.33}$$

Consequently $G_s(z) = (z, h_s(z))$ maps the closed unit disc into U.

Suppose $s = 0$ or $s = s_j$. For $|\zeta| = 1$, equations (2.10)–(2.12) give $u = s$ and $Y = 0$ at $(\zeta, h_s(\zeta))$. Since $Q(s) = 0$,

$$\rho(\zeta, h_s(\zeta)) = 0.$$

At these points, (2.33) gives

$$b(\zeta, h_s(\zeta)) \leq \frac{1}{10} + \frac{16}{r^2}(1.01\sigma)^2 - 4 = -2.8799.$$

Therefore $R = 0$ on the boundary of this graph disc. By (2.27), $R \circ G_s$ is strictly subharmonic: the first component of G_s' is always one. The maximum principle now gives

$$R(G_s(z)) < 0 \qquad (z \in \Delta). \tag{2.34}$$

Thus every disc in (2.1) is a section of π.

Conversely, let

$$G(z) = (z, g(z)) : \Delta \longrightarrow \Omega$$

be any holomorphic section. By (2.29), g is bounded. Fatou's theorem supplies radial limits $g^*(\zeta)$ for almost every $\zeta \in \partial\Delta$. At such a point,

$$(\zeta, g^*(\zeta)) \in \overline{\Omega}.$$

Using $R \geq \rho$ and (2.15), we obtain

$$0 \geq R(\zeta, g^*(\zeta)) \geq 2Y(\zeta, g^*(\zeta))^2 + Q(u(\zeta, g^*(\zeta))) \geq 0.$$

Both nonnegative summands therefore vanish. Equations (2.5) and (2.12) imply that, almost everywhere,

$$g^*(\zeta) = h_s(\zeta)$$

for some

$$s \in S := \big(\{0\} \cup \{\sigma 2^k : k \in \mathbb{Z}\}\big) \cap (-2r, 2r). \tag{2.35}$$

Radial boundary values are measurable, so each set

$$E_s = \{\zeta : g^*(\zeta) = h_s(\zeta)\}$$

is measurable. Since the countable union over $s \in S$ has full measure, at least one E_s has positive measure.

We recall the short boundary-uniqueness argument needed here. A bounded holomorphic function H whose radial limit vanishes on a set of positive measure must be identically zero. Otherwise, remove its finite-order zero at the origin and normalize the resulting bounded function so that $|H| \leq 1$ and $H(0) \neq 0$. Jensen's inequality gives

$$\frac{1}{2\pi}\int_0^{2\pi} -\log|H(te^{i\theta})|\,d\theta \leq -\log|H(0)|, \qquad 0 < t < 1.$$

Fatou's lemma contradicts this uniform bound, because the integrand tends to $+\infty$ on a set of positive measure.

Applying this argument to $g - h_s$, we conclude that

$$g \equiv h_s \quad \text{for some } s \in S. \tag{2.36}$$

The cap excludes precisely the extra positive dyadic parameters. Indeed, if $s = \sigma 2^k$ with $k \geq 1$, then $s \geq 2\sigma = r/2$, whereas

$$\max_{|\zeta|=1} |h_s(\zeta)| = |A(s)| + |B(s)| > |A(s)| \geq s \geq r/2.$$

This contradicts $|g| \leq r/2$ on the closed disc, which follows from (2.29) and the continuity of the affine function $g = h_s$. Together with (2.34), this proves the exact classification:

$$\boxed{\text{Every holomorphic section of } \pi \text{ is } G_* \text{ or } G_j \text{ for some } j \geq 1.} \tag{2.37}$$

2.5. **Intersections and exact minimizing chains.** It remains to examine how these sections intersect.

For distinct positive s, t, the graphs of h_s and h_t intersect at the unique base point

$$z_{s,t} = -\eta \, \frac{(s+t) + \frac{9i}{7}\eta(s^2+st+t^2)}{1+i\eta(s+t)}. \tag{2.38}$$

Taking the imaginary part gives

$$\operatorname{Im} z_{s,t} = -\frac{\eta^2(2s-t)(s-2t)}{7(1+\eta^2(s+t)^2)}. \tag{2.39}$$

Thus two distinct positive dyadic sections intersect over the real axis exactly when their parameters are adjacent in the dyadic sequence. For the sections in (2.1), these intersections occur at

$$\xi_j = z_{s_j,s_{j+1}} = -\eta(s_j+s_{j+1}) = -\frac{3}{2}\eta s_j, \qquad \xi_j \uparrow 0. \tag{2.40}$$

A nonzero section meets the zero graph at

$$z_{s,0} = -\eta s \, \frac{1+\frac{9i}{7}\eta s}{1+i\eta s},$$

and

$$\operatorname{Im} z_{s,0} = -\frac{2\eta^2 s^2}{7(1+\eta^2 s^2)} < 0. \tag{2.41}$$

Consequently, *no nonzero section meets the zero section over the real axis.*

Define

$$a = -2\eta\sigma, \qquad p = (a, h_\sigma(a)), \qquad q_N = (0, B(s_N)), \qquad q_* = (0,0). \tag{2.42}$$

Explicitly,

$$h_\sigma(a) = -\eta\sigma^2 - \frac{5i}{7}\eta^2\sigma^3.$$

Thus $\|p\| < 3\eta\sigma < \sigma$, and

$$\|q_N\| \le 2\eta\sigma^2 < \sigma.$$

All these points belong to the compact set C in (2.32), and $q_N \to q_*$.

We next compute their Kobayashi distances. Apply Schwarz–Pick to the projection of each link of a chain, and then the Poincaré triangle inequality. Taking the infimum over all chains gives

$$k_\Omega(P,Q) \ge \rho_\Delta(\pi P, \pi Q). \tag{2.43}$$

Thus projection supplies a common lower bound for the cost of every chain.

For each N, follow G_1 from a to ξ_1, then G_2 from ξ_1 to ξ_2, and so forth, finishing on G_N from ξ_{N-1} to 0. For $N = 1$, simply follow G_1 from a to 0. Because

$$a < \xi_1 < \xi_2 < \cdots < 0,$$

the costs telescope along a real Poincaré geodesic:

$$\rho_\Delta(a,\xi_1)+\cdots+\rho_\Delta(\xi_{N-1},0)=\rho_\Delta(a,0).$$

Together with (2.43), this proves

$$k_\Omega(p,q_N)=\rho_\Delta(a,0)=\operatorname{artanh}(2\eta\sigma). \tag{2.44}$$

The vertical disc

$$\zeta\longmapsto(0,\sigma\zeta)$$

lies in Ω, by (2.32). Appending a link in this disc from q_N to q_* costs

$$\operatorname{artanh}\bigl(|B(s_N)|/\sigma\bigr)\longrightarrow 0.$$

Applying (2.43) again gives

$$k_\Omega(p,q_*)=\rho_\Delta(a,0). \tag{2.45}$$

We now describe the rigidity of any chain realizing one of these distances. Suppose a chain has discs $f_i:\Delta\to\Omega$, marked points α_i,β_i, and junctions $P_0,\dots,P_m$. Write $z_i=\pi(P_i)$. For endpoints as in (2.44) or (2.45),

$$\begin{aligned}\rho_\Delta(a,0)&\le\sum_{i=1}^{m}\rho_\Delta(z_{i-1},z_i)\\&\le\sum_{i=1}^{m}\rho_\Delta(\alpha_i,\beta_i)=k_\Omega(P_0,P_m)=\rho_\Delta(a,0).\end{aligned} \tag{2.46}$$

Every inequality is therefore an equality.

Discard zero-cost links. Equality in Schwarz–Pick implies that each

$$\pi\circ f_i:\Delta\longrightarrow\Delta$$

is an automorphism. Reparametrizing f_i by its inverse makes it a section of π. By (2.37), every remaining disc is one of $G_*,G_1,G_2,\dots$.

Equality in the Poincaré triangle inequality also places all projected junctions, in order, on the real interval $[a,0]$. Indeed, equality forces each junction onto the unique Poincaré geodesic segment joining the endpoints.

For a chain from p to q_N, its first section must be G_1. To verify uniqueness, (2.39) shows that the only other positive section that could meet G_1 over the real axis is G_2, and its intersection lies at

$$\xi_1=-\frac{3}{2}\eta\sigma\neq a.$$

Equation (2.41) excludes G_*.

Its final section must be G_N, since

$$h_s(0)=B(s),\qquad \operatorname{Re}B(s)=\eta s^2,$$

and this real part determines the nonnegative parameter s uniquely.

At each change of nonzero section, (2.39) allows only adjacent indices. The zero section cannot intervene, by (2.41). A sequence starting at index 1 and ending at index N, changing only between adjacent indices, requires

at least N discs. The chain already constructed has exactly N, proving the first assertion.

For a chain from p to q_*, the first section must again be G_1, while the last must be G_*, since $B(s) \neq 0$ for every positive s. There would therefore have to be a transition from a nonzero section to the zero section over a real projected junction. Equation (2.41) makes that impossible. This proves the second assertion.

2.6. **Attainment for a fixed number of discs.** Finally, we justify the strict inequality for every fixed number of discs. This is a special case of [14, Proposition 4(b)]; we give the argument directly. For our fixed endpoints, the infimum over chains with at most m discs is attained.

To prove this, pad chains with constant discs so that every chain has exactly m links. In a minimizing sequence, normalize each disc's marked points to

$$0,\ t_{i,\nu}, \qquad 0 \le t_{i,\nu} < 1.$$

If the total costs are bounded by L, then

$$0 \le t_{i,\nu} \le \tanh L < 1.$$

Boundedness of Ω, Montel's theorem, and passage to a subsequence give simultaneous limits

$$t_{i,\nu} \to t_i, \qquad f_{i,\nu} \to f_i$$

locally uniformly, with $f_i : \Delta \to \overline{\Omega}$.

By (2.29), $\overline{\Omega} \Subset U$, so R is defined and plurisubharmonic on a neighborhood of every limit image. The first limit disc starts at $p \in \Omega$. Since $R \circ f_1 \le 0$ is subharmonic, it cannot attain zero at an interior point without being identically zero, contradicting $R(p) < 0$. Hence $f_1(\Delta) \subset \Omega$. Its endpoint is interior, so the same argument applies to f_2, and inductively to every limit disc. The junction relations pass to the limit because the parameters remain in a compact subdisc. The costs also converge, since artanh is continuous there.

Thus the fixed-m infimum is attained. It is finite for our endpoints, which lie in an interior Euclidean ball. If it equalled $k_\Omega(p, q_*)$, its minimizer would be a finite realizing chain, contrary to what we proved. Therefore

$$k_\Omega^{(m)}(p, q_*) > k_\Omega(p, q_*) \qquad (m < \infty).$$

□

2.7. **Exponential errors on the entire obstruction family.** We now prove Corollary 1.2. The argument also explains why the exact obstruction cannot establish the optimality of the general rate in Theorem 1.4.

Proof of Corollary 1.2. By (2.44)–(2.45), all the distances under consideration equal

$$D = \operatorname{arctanh}(2\eta\sigma).$$

The construction above supplies an actual m-disc chain from p to q_m with cost exactly D. By (2.32), the entire vertical disc

$$v(\zeta) = (0, \sigma\zeta), \qquad \zeta \in \Delta, \tag{2.47}$$

lies in Ω. We will use it to pass from q_m to any later endpoint, or to q_*.

For $0 < s \le \sigma$, the explicit formula for B gives

$$|B(s)| = \eta s^2 \sqrt{1 + \frac{81}{49}\eta^2 s^2} \le 2\eta s^2.$$

Consequently, with $\alpha_m = B(s_m)/\sigma$,

$$|\alpha_m| \le 2\eta\sigma\, 4^{1-m} = 8\eta\sigma\, 4^{-m} < \frac{1}{4}. \tag{2.48}$$

If $q = q_N$ with $N \le m$, its exact N-disc chain already has the required cost. If $N > m$, put $\beta = B(s_N)/\sigma$; for $q = q_*$ put $\beta = 0$. Both the real and the imaginary parts of $B(s)$ are nonnegative and increasing for $s \ge 0$. For $N > m$ we have $s_N < s_m$, and the case $\beta = 0$ is immediate. In either case,

$$|\alpha_m - \beta| \le |\alpha_m|, \qquad |\beta| \le |\alpha_m| < \frac{1}{4}.$$

Thus the pseudohyperbolic separation of the two marked points on (2.47) satisfies

$$t := \frac{|\alpha_m - \beta|}{|1 - \overline{\alpha_m}\beta|} \le \frac{16}{15}|\alpha_m| < \frac{4}{15}.$$

For $0 \le t \le 4/15$, integration of $(1 - t^2)^{-1}$ gives $\operatorname{arctanh} t \le 2t$. Therefore

$$\rho_\Delta(\alpha_m, \beta) \le \frac{32}{15}|\alpha_m| \le 4|\alpha_m| \le C_{\log} 4^{-m}. \tag{2.49}$$

Append this one vertical link to the exact chain ending at q_m. The result is an actual chain with at most $m + 1$ discs and cost at most $D + C_{\log}4^{-m}$, uniformly over all $q \in \mathcal{Q}$. This proves (1.1).

For the strict tolerance assertion, set

$$L = \max\left\{0, \left\lceil \log_4 \frac{C_{\log}}{\varepsilon} \right\rceil\right\}, \qquad m = L + 1.$$

Then $C_{\log}4^{-L} \le \varepsilon$, including when $L = 0$. The chain just constructed has excess at most

$$C_{\log}4^{-m} \le \frac{\varepsilon}{4} < \varepsilon$$

and uses at most $m + 1 = L + 2$ discs. This is (1.2). □

Remark 2.1. For the limiting pair, Theorem 1.1 and (1.1) give

$$0 < k_\Omega^{(m+1)}(p, q_*) - k_\Omega(p, q_*) \le C_{\log}4^{-m} \qquad (m \ge 1).$$

Hence strictly positive errors at every finite chain number coexist with an exponential upper bound on those errors. Even if N is allowed to depend on ε, the family (p, q_N) cannot yield a positive-power lower bound for the number of discs needed at tolerance ε. This does not exclude slower approximation

for other endpoint pairs in the same domain. The next subsection improves the logarithmic bound and proves that it is not sharp.

2.8. **A sharper bound from the flat boundary term.** The vertical correction in Corollary 1.2 is not optimal. We instead construct holomorphic graphs whose boundary values satisfy $Y = 0$ exactly, without imposing the discrete condition $Q = 0$. The remaining boundary error is then controlled by the flatness of Q. The construction below uses only the explicit polynomial boundary relation; no real-analyticity of $\partial\Omega$ is assumed.

Lemma 2.2 (Auxiliary graphs and real intersections)**.** *There exists $s_0 > 0$ such that, for every $0 < s < s_0$ and every $0 \le \beta \le 1/2$, there are a function $g = g_{s,\beta}$ holomorphic on a neighborhood of $\overline{\Delta}$, a real function u on $\partial\Delta$, and a real point $x = x_{s,\beta}$ with*

$$g(0) = B(s\beta), \qquad g(x) = h_s(x), \qquad -2\eta s < x < -\eta s, \tag{2.50}$$

$$\begin{gathered} g(\zeta) = A(u(\zeta))\zeta + B(u(\zeta)) \quad (|\zeta| = 1), \\ |u| \le 3s, \qquad \|g\|_{\overline{\Delta}} \le 7s. \end{gathered} \tag{2.51}$$

Proof. Let $\mathcal{A}_2$ be the Banach algebra of Laurent series

$$f(\zeta) = \sum_{k\in\mathbb{Z}} f_k \zeta^k, \qquad \|f\|_2 = \sum_{k\in\mathbb{Z}} |f_k|\, 2^{|k|} < \infty.$$

Multiplication is bounded with constant one, since $2^{|j+k|} \le 2^{|j|}2^{|k|}$. Conjugation on the unit circle is an isometry. Let $P_{\le 0}$ retain the nonpositive Fourier coefficients, and define the real-linear operator

$$\mathcal{L}f = \zeta^{-1}P_{\le 0}f + \overline{\zeta^{-1}P_{\le 0}f}.$$

Then $\|\mathcal{L}\| \le 4$, its values are real on the circle, and its constant coefficient is zero.

For real parameters λ, T, β, put

$$a_\lambda = \lambda\beta^2 + \frac{9i}{7}\lambda^2\beta^3, \qquad N_\lambda(U) = \lambda(1+i\zeta)U^2 + \frac{9i}{7}\lambda^2 U^3.$$

We solve in the real subspace of $\mathcal{A}_2$ the equation

$$U = T + a_\lambda\zeta^{-1} + \overline{a_\lambda}\zeta - \mathcal{L}N_\lambda(U). \tag{2.52}$$

For $|T| \le 2$, $|\beta| \le 1$, and $|\lambda| \le 10^{-4}$, the free term on the right has norm less than 2.001. In the closed ball of radius $1/4$ about that free term, $\|U\|_2 < 3$. The norm of the correction and its Lipschitz constant there are bounded, respectively, by

$$4\left(27|\lambda| + \frac{243}{7}\lambda^2\right) < 0.011, \qquad 4\left(18|\lambda| + \frac{243}{7}\lambda^2\right) < 0.008.$$

The contraction theorem therefore gives a unique fixed point in this ball. The fixed point is real analytic in the real parameters locally around the compact parameter ranges used below. Indeed, the equation is polynomial in these parameters and in U, and its derivative in U is the identity plus

an operator of norm less than 0.008; its inverse is given by a convergent Neumann series. In particular, parameter derivatives are uniformly bounded on smaller compact ranges.

Set

$$H_{\lambda,T,\beta} = \zeta U + N_\lambda(U).$$

Multiplying (2.52) by ζ shows that all negative Fourier coefficients of H vanish and its constant coefficient is exactly a_λ. More explicitly, the nonpositive part of ζU is $a_\lambda - P_{\le 0} N_\lambda(U)$, since the conjugate term, after multiplication by ζ, has only powers at least two. Thus H is holomorphic for $|z| < 2$, with

$$H(0) = a_\lambda, \qquad \|H\|_2 < 7. \tag{2.53}$$

We next arrange a real intersection. Differentiating (2.52) at $\lambda = 0$ gives

$$U = T + \lambda(\beta^2 - T^2)(\zeta + \zeta^{-1}) + O_{\mathcal{A}_2}(\lambda^2).$$

Consequently,

$$H(z) = a_\lambda + (T + i\lambda T^2)z + \lambda(\beta^2 - T^2)z^2 + \lambda^2 z E(\lambda, T, \beta; z). \tag{2.54}$$

Here E is holomorphic in z and real analytic in the parameters, with uniform bounds, including first parameter derivatives, near $T \in [2/7, 1/2]$, $\beta \in [0, 1/2]$, $\lambda = 0$, and $|z| \le 1$. The factor z in the remainder follows from the exact equality $H(0) = a_\lambda$; analyticity in λ and the computed first derivative give the factor λ^2. These facts also follow by dividing the corresponding convergent power series.

Take $\lambda = \eta s$ and seek an intersection at $x = -c\lambda$. Since

$$\frac{h_s(-c\lambda)}{s} = \lambda(1 - c) + i\lambda^2 \left(\frac{9}{7} - c\right),$$

define, initially for $\lambda \neq 0$,

$$F_1(\lambda, T, c, \beta) = \lambda^{-1} \operatorname{Re}\left[H(-c\lambda) - \lambda(1 - c) - i\lambda^2 \left(\frac{9}{7} - c\right)\right],$$
$$F_2(\lambda, T, c, \beta) = \lambda^{-2} \operatorname{Im}\left[H(-c\lambda) - \lambda(1 - c) - i\lambda^2 \left(\frac{9}{7} - c\right)\right].$$

Equation (2.54) shows that both functions have real-analytic extensions across $\lambda = 0$. In fact, the bracket equals

$$\begin{aligned} &\lambda[\beta^2 - 1 + c(1 - T)] \\ &\quad + i\lambda^2 \left[\frac{9}{7}(\beta^3 - 1) + c(1 - T^2)\right] \\ &\quad + \lambda^3[c^2(\beta^2 - T^2) - cE(\lambda, T, \beta; -c\lambda)]. \end{aligned}$$

At $\lambda = 0$ the simultaneous roots are

$$T_\beta = \frac{9}{7}\frac{1 - \beta^3}{1 - \beta^2} - 1 = \frac{2 + 2\beta + 9\beta^2}{7(1 + \beta)}, \qquad c_\beta = \frac{1 - \beta^2}{1 - T_\beta}. \tag{2.55}$$

For $0 \le \beta \le 1/2$, elementary simplification gives

$$\frac{2}{7} \le T_\beta \le \frac{1}{2}, \qquad \frac{7}{5} \le c_\beta \le \frac{3}{2}.$$

The determinant of the Jacobian in (T, c) is

$$\det \frac{\partial(F_1, F_2)}{\partial(T, c)}\Big|_{(0,T_\beta,c_\beta,\beta)} = -c_\beta(1 - T_\beta)^2,$$

whose absolute value is at least $7/20$. The implicit function theorem, uniformly on the compact interval $0 \le \beta \le 1/2$, supplies a common $\lambda_0 > 0$ and exact roots $T(\lambda, \beta), c(\lambda, \beta)$ for $|\lambda| < \lambda_0$. Shrinking λ_0, we may require

$$\frac{1}{5} < T(\lambda, \beta) < \frac{3}{5}, \qquad 1 < c(\lambda, \beta) < 2.$$

Uniformity follows by a finite cover of the compact curve of roots in (2.55); the locally unique branches agree on overlaps after reducing the common parameter neighborhood.

For $s > 0$ with $\eta s < \min(\lambda_0, 10^{-4})$, define

$$g_{s,\beta}(z) = sH_{\eta s, T(\eta s,\beta),\beta}(z), \qquad x = -c(\eta s, \beta)\eta s.$$

The two equations $F_1 = F_2 = 0$ prove (2.50). On the unit circle set $u = sU$, which is real and satisfies $|u| \le 3s$. The definition of H gives the exact boundary identity in (2.51); its norm bound follows from (2.53). This proves the lemma. □

Proof of Theorem 1.3. Choose $m_0 \ge 2$ so large that, for every $m \ge m_0$, $s = s_m$ satisfies the preceding lemma and

$$7s \le r/4.$$

This choice is possible because $s_m \to 0$. It includes the possibly smaller implicit-function threshold; the numerical value of η alone is not used to assert that threshold.

If $q = q_N$ with $N \le m$, use its exact chain. Otherwise put $\beta = s_N/s_m \in (0, 1/2]$ for $q = q_N$, and put $\beta = 0$ for $q = q_*$. Apply Lemma 2.2, and write $g = g_{s,\beta}$ and $G(z) = (z, g(z))$. This graph maps a neighborhood of $\overline{\Delta}$ into U after a harmless restriction of that neighborhood. On $\partial\Delta$, the real parameter in (2.10) agrees with the u in (2.51), by uniqueness: $|u| \le 3s < 2r$ and $|g| \le r/4$. Thus $Y(\zeta, g(\zeta)) = 0$ and

$$\rho(G(\zeta)) = Q(u(\zeta)), \qquad b(G(\zeta)) \le \frac{1}{10} + 1 - 4 = -2.9.$$

Since $Q \ge 0$, the cap is inactive by (2.24), so $R(G(\zeta)) = Q(u(\zeta))$. Both branches of (2.4), including the value at zero, give

$$R(G(\zeta)) \le \mu_s := \eta \exp\left(-\frac{1}{9s^2}\right) \qquad (|\zeta| = 1). \tag{2.56}$$

The global Levi bound (2.27) applies on the entire graph in U, even where the graph has not yet been shown to lie in Ω. Since the first component of

G' equals one, $R \circ G - |z|^2/10$ is subharmonic. Its boundary values are at most $\mu_s - 1/10$. The maximum principle yields

$$R(G(z)) \le \mu_s - \frac{1}{10}(1 - |z|^2) \qquad (z \in \overline{\Delta}). \tag{2.57}$$

Let

$$r_s = \sqrt{1 - 20\mu_s}.$$

Here $0 < \mu_s \le \eta < 1/80$, so $r_s > 3/4$. Equation (2.57) shows that

$$f_{s,\beta}(\zeta) = G(r_s\zeta)$$

maps the entire closed unit disc into Ω, with $R \le -\mu_s$. It contains q at parameter zero and the intersection point $G_m(x) = G(x)$ at parameter x/r_s.

The preceding exact junction is $\xi_{m-1} = -3\eta s_m$. By (2.50), $\xi_{m-1} < x < 0$. Follow $G_1, \dots, G_m$ as before, stopping at x on the last graph, then use $f_{s,\beta}$ to reach q. The prefix uses m discs and costs

$$\rho_\Delta(a, x) = D - \operatorname{arctanh}|x|, \qquad D = \operatorname{arctanh}(2\eta\sigma).$$

All marked parameters of the last disc belong to Δ, since $|x| \le 2\eta s < 1/8$ and $r_s > 3/4$. Moreover,

$$\frac{1}{r_s} - 1 = \frac{20\mu_s}{r_s(1 + r_s)} < 16\mu_s.$$

The derivative of $\operatorname{arctanh} t$ is at most two between $|x|$ and $|x|/r_s$. Hence the excess cost of this actual $(m+1)$-disc chain is at most

$$\begin{aligned} \operatorname{arctanh}\frac{|x|}{r_s} - \operatorname{arctanh}|x| &\le 32|x|\mu_s \\ &\le 64\eta^2 s \exp\left(-\frac{1}{9s^2}\right). \end{aligned} \tag{2.58}$$

Since $s = s_m = \sigma 2^{1-m} \le \sigma$, the last expression is at most

$$64\eta^2\sigma \exp\left(-\frac{4^m}{36\sigma^2}\right).$$

This proves the asserted actual-chain estimate and (1.3), uniformly for all $q \in \mathcal{Q}$.

For any $\varepsilon > 0$, choose m to be the maximum inside (1.4). Then

$$c_{\mathrm{flat}} 4^m \ge \log\left(1 + \frac{2C_{\mathrm{flat}}}{\varepsilon}\right),$$

so the excess is at most

$$\frac{C_{\mathrm{flat}}}{1 + 2C_{\mathrm{flat}}/\varepsilon} < \frac{\varepsilon}{2} < \varepsilon.$$

The chain uses at most $m + 1$ discs, proving (1.4). □

Remark 2.3 (What remains of the lower-bound question). Let $\mathcal{N}_{\mathcal{Q}}(\varepsilon)$ denote the least uniform number of discs needed for strict ε-approximation on $\{p\} \times \mathcal{Q}$. Theorem 1.3 proves that the earlier logarithmic upper bound is not sharp, since

$$\mathcal{N}_{\mathcal{Q}}(\varepsilon) = O(\log\log(1/\varepsilon)) = o(\log(1/\varepsilon)).$$

On the other hand, $\mathcal{N}_{\mathcal{Q}}(\varepsilon) \to \infty$ as $\varepsilon \downarrow 0$. Indeed, for every fixed M, Theorem 1.1 gives the positive gap

$$d_M = k_\Omega^{(M)}(p, q_*) - k_\Omega(p, q_*) > 0.$$

For $0 < \varepsilon < d_M$, no M-disc chain can give the required approximation even for q_*. This is qualitative divergence; it supplies no rate of decay for d_M. A matching lower bound for the double-logarithmic count, or a still faster upper bound, remains open here.

2.9. **A precise obstruction to lossless merging.** The following consequence specifies the uniform assertion that fails. It is stronger than merely observing an unbounded number of discs in some minimizing chains.

Corollary 2.4. *For the domain constructed above, set $b = 1/3$ and $c = 2/3$. There are discs $F_j, H_j \in \mathcal{O}(\Delta, \Omega)$ such that*

$$\|F_j - H_j\|_\infty \longrightarrow 0, \qquad F_j(b) = H_j(b),$$

but, for all sufficiently large j, no disc $f \in \mathcal{O}(\Delta, \Omega)$ satisfies $f(0) = F_j(0)$ and $f(c) = H_j(c)$. The two-disc chains through their common marked points are exactly minimizing.

Proof. Use the sections and intersection parameters from (2.1) and (2.40). Put

$$a_j = \frac{\xi_j - b}{1 - b\xi_j}, \qquad T_j(\zeta) = \frac{\zeta + a_j}{1 + a_j\zeta}, \qquad F_j = G_j \circ T_j, \quad H_j = G_{j+1} \circ T_j.$$

Then $T_j(b) = \xi_j$, and the equality at b follows. Since $s_j \to 0$, the explicit affine formulas imply

$$\|F_j - H_j\|_\infty = \|G_j - G_{j+1}\|_\infty \longrightarrow 0.$$

If the asserted interpolating disc f existed, $T_j^{-1} \circ \pi \circ f$ would fix 0 and c, and hence would be the identity by Schwarz's lemma. Thus $f \circ T_j^{-1}$ would be a section of π. But

$$T_j(0) \longrightarrow -\frac{1}{3}, \qquad T_j(c) \longrightarrow \frac{3}{7}.$$

For large j these base points are distinct from every real intersection parameter in (2.40). The first endpoint therefore forces this section to be G_j, while the second forces it to be G_{j+1}, a contradiction. Finally, projection bounds the endpoint distance below by $\rho_\Delta(0, c)$, and the two-disc chain has exactly that cost, since $0 < b < c$ lie on one Poincaré geodesic. □

Thus no positive closeness threshold depending only on Ω, b, c guarantees this normalized lossless interpolation. The corollary addresses this precise uniform assertion; it does not rule out a construction with positive error or with additional hypotheses on the discs. In particular, it does not contradict the literal fixed-positive-error assertion of [9, Theorem 18, pp. 12–13].

3. Uniform approximation by short chains

We prove Theorem 1.4. All constants in this section are fixed before the endpoints and the tolerance are chosen. Complex affine rescaling does not change chain costs, so assume

$$\operatorname{diam}\Omega \le 1.$$

Write

$$\delta(x) = \operatorname{dist}(x, \partial\Omega).$$

If $n = 1$, let $p : \Delta \to \Omega$ be a universal covering and fix $\alpha \in p^{-1}(z)$. Lifting successive discs of any finite chain starting at α, and applying Schwarz–Pick upstairs, gives

$$k_\Omega(z, w) = \inf_{\beta \in p^{-1}(w)} \rho_\Delta(\alpha, \beta).$$

The reverse inequality follows by using p itself as a disc. Every tolerance is therefore achieved with one disc. Henceforth $n \ge 2$.

3.1. **Fixed metric constants and geodesic estimates.** Fix $o \in \Omega$. Choose $A_r, H \ge 0$ and $0 < c \le 1$ such that

$$\left| k_\Omega(o, x) - \frac{1}{2}\log\frac{1}{\delta(x)} \right| \le A_r, \qquad \kappa_\Omega(x; v) \ge \frac{c|v|}{\sqrt{\delta(x)}}, \tag{3.1}$$

and all real Kobayashi geodesic triangles are H-thin.

These estimates follow from [1, Proposition 1.2, Corollary 1.3, and Theorem 1.4], with the radial estimate obtained from their distance comparison. All require only C^2 boundary.

Set

$$E = e^{4A_r}, \qquad W = e^{4A_r + 8H}, \qquad b = e^{-4A_r - 2H},$$

$$V = \frac{2\sqrt{W}}{c}, \qquad Q_0 = 4W^{1/4}. \tag{3.2}$$

The radial estimate and the triangle inequality give

$$E^{-1}\delta(x)e^{-2k_\Omega(x,y)} \le \delta(y) \le E\delta(x)e^{2k_\Omega(x,y)}. \tag{3.3}$$

Consequently, for every holomorphic $f : \Delta \to \Omega$,

$$\delta(f(\zeta)) \ge \frac{\delta(f(0))}{E}\frac{1 - |\zeta|}{1 + |\zeta|}. \tag{3.4}$$

We record why real geodesics and their required Euclidean estimates are available without identifying κ_Ω with the metric derivative of k_Ω.

A centered disc has derivative of norm at most 1, by Cauchy's estimate and the diameter normalization. Thus a disc-chain path, parametrized by Poincaré arclength, has Euclidean speed at most 1. In particular,

$$|x-y| \le k_\Omega(x,y). \tag{3.5}$$

Local affine discs give the Euclidean topology. Estimate (3.1) makes closed Kobayashi balls compact, while disc-chain paths show that k_Ω is a length metric. The usual compactness argument for near-minimizing paths therefore produces real geodesics.

More precisely, a chain of cost S beginning at x has depths at most $\delta(x)+S$. Using (3.1) along its disc arcs gives

$$|x-y| \le \frac{S\sqrt{\delta(x)+S}}{c}.$$

Letting $S \downarrow k_\Omega(x,y)$, then applying this to short subsegments of a unit-speed real geodesic γ, yields

$$|\gamma'(s)| \le \frac{\sqrt{\delta(\gamma(s))}}{c} \quad \text{for almost every } s. \tag{3.6}$$

Choose $p=\gamma(0)$ minimizing $k_\Omega(o,\cdot)$ on a geodesic segment. Thin triangles imply

$$k_\Omega(o,p)+|s|-4H \le k_\Omega(o,\gamma(s)) \le k_\Omega(o,p)+|s|. \tag{3.7}$$

For completeness, a point on $[p,\gamma(s)]$ at distance $t>2H$ from p cannot be H-close to $[o,p]$: that would give a point of the original segment closer to o than p. It is therefore H-close to $[o,\gamma(s)]$, giving the lower bound with error $t+2H$; let $t \downarrow 2H$. Shorter segments satisfy the same inequality directly.

Combining (3.1) and (3.7),

$$E^{-1}\delta(p)e^{-2|s|} \le \delta(\gamma(s)) \le W\delta(p)e^{-2|s|}. \tag{3.8}$$

Hence, if the segment has maximum depth at most d,

$$\operatorname{length}_{\mathrm{Eucl}}(\gamma) \le V\sqrt{d}, \qquad \int_\gamma \delta^2\,ds \le \frac{W^2}{2}d^2. \tag{3.9}$$

For every segment, since $\delta \le 1$,

$$\int_\gamma \delta^{1/4}\,ds \le Q_0. \tag{3.10}$$

3.2. **The C^2 convexification and localization inputs.** We use two consequences of local convexification.

First, there are finitely many neighborhoods

$$W_j \Subset V_j \Subset U_j, \qquad \partial\Omega \subset \bigcup_j W_j,$$

such that $\Omega\cap U_j$ is biholomorphic to a convex domain, and a common $L \ge 1$ satisfies

$$\kappa_{\Omega\cap U_j}(x;v) \le (1+L\delta(x)^2)\gamma_\Omega(x;v), \qquad x \in \Omega\cap V_j. \tag{3.11}$$

Here γ_Ω denotes the infinitesimal Carathéodory metric.

Second, near each boundary point there is a holomorphic map, defined on all of Ω and across a boundary neighborhood,

$$\Phi_i : \Omega \cup \mathcal{U}_i \longrightarrow \mathbb{C}^n, \qquad \Phi_i(\Omega) \subset D_i, \tag{3.12}$$

where D_i is bounded, C^2, and strongly convex. The map is biholomorphic near the chosen boundary point, with inverse Ψ_i, and the domains agree there:

$$\Phi_i(\Omega \cap \mathcal{U}_i) = D_i \cap \Phi_i(\mathcal{U}_i). \tag{3.13}$$

The localization estimate (3.11) is [12, Proposition 2, equation 8]. Its Stein-compact alternative applies: a C^2 strongly pseudoconvex boundary has a strictly plurisubharmonic two-sided defining function in a collar, and small outward levels give Stein neighborhoods of $\overline{\Omega}$. The maps (3.12)–(3.13) are supplied by the localized Fornæss construction [4, Proposition 1], as used in the proof of [12, Proposition 2]; the C^2 strongly convex target is explicit in [5, Lemma 3.5, p. 60]. The map Φ_i is used globally only in the forward direction. Its inverse Ψ_i is used exclusively on the fixed local chart where (3.13) holds.

Choose each inverse chart on a neighborhood of

$$\overline{B(\xi_i, 8r_i)}, \qquad \xi_i \in \partial D_i, \quad r_i > 0.$$

After affine rescaling, assume $\operatorname{diam} D_i \le 1$. Choose compact boundary patches P_i covering $\partial\Omega$, with

$$P_i \Subset \mathcal{U}_i, \qquad \Phi_i(P_i) \subset B(\xi_i, r_i).$$

Shrink the patches if necessary so that uniform boundary-distance comparisons hold near them.

Proposition 3 of [12], applied to the convex target, supplies a constant M_i^{loc} such that

$$\kappa_{D_i \cap B(\xi_i, 4r_i)}(x; v) \le (1 + M_i^{\mathrm{loc}} \delta_{D_i}(x)^2) \kappa_{D_i}(x; v), \quad x \in D_i \cap B(\xi_i, 3r_i). \tag{3.14}$$

Its hypotheses are boundedness, convexity, visibility, and Dini-smooth boundary; C^2 strong convexity satisfies them.

Fix $M \ge 2$ dominating:

- the forward and inverse chart derivative bounds on these fixed neighborhoods;
- the boundary-distance comparison factors;
- all M_i^{loc};
- constants in the half-disc oscillation estimates

$$\sup_{|\zeta| \le 1/2} |f(\zeta) - f(0)| \le M \sqrt{\delta_G(f(0))} \tag{3.15}$$

for $G = \Omega, D_1, \dots, D_N$ and $f : \Delta \to G$.

The last constants follow by integrating the infinitesimal lower bounds, using (3.3) for Ω. For a convex target one can use the sharper elementary inequality

$$\delta_D(f(0))\frac{1-r}{1+r} \le \delta_D(f(\zeta)) \le \delta_D(f(0))\frac{1+r}{1-r}, \qquad |\zeta| \le r. \tag{3.16}$$

To prove (3.16), express δ_D as the infimum of distances to supporting hyperplanes and apply harmonic Harnack to each positive affine support function composed with f.

Choose $0 < \beta \le 1$ smaller than every r_i, every source-chart buffer around P_i, and a neighborhood width on which

$$M^{-1}\delta(x) \le \delta_{D_i}(\Phi_i x) \le M\delta(x) \tag{3.17}$$

holds near P_i. Set

$$a_0 = \left(\frac{\beta}{64M^2}\right)^2. \tag{3.18}$$

Here is the resulting containment fact, used below. If

$$a = \delta(z) \le a_0, \qquad S \le a/16,$$

and a chain of cost S starts at z, choose i with a nearest boundary point of z in P_i. Its chain centers remain within S of z. Estimate (3.15) keeps all their centered source half-discs inside $\mathcal{U}_i$. Their target half-discs lie in $B(\xi_i, 2r_i)$. The same is true for half-discs obtained by recentering a target geodesic based at $\Phi_i z$ by source distances at most S.

Indeed, the relevant source displacements are bounded by

$$2a + M\sqrt{2a},$$

and the relevant target displacements, in addition to the initial radius r_i, are bounded by

$$4M^2\sqrt{a} \le \beta/16.$$

Thus chart containment is established before any rigidity estimate is applied.

3.3. **A stationary-disc estimate allowing different centers.** Let D be a bounded C^2 strongly convex domain and $K \Subset D$. Fix $m_D > 0$ such that

$$\mathrm{Re}\big[(\xi - v)\cdot\overline{\nu_D(\xi)}\big] \ge m_D|\xi - v|^2, \qquad \xi \in \partial D, \quad v \in \overline{D}, \tag{3.19}$$

where the dot is complex bilinear and ν_D is the outward unit normal.

A complex geodesic $g : \Delta \to D$ has a holomorphic dual $\widetilde{g}$, continuous on $\overline{\Delta}$, normalized by

$$g' \cdot \widetilde{g} = 1, \qquad \widetilde{g}(\zeta) = \zeta\mu_g(\zeta)\overline{\nu_D(g(\zeta))} \quad (\zeta \in \partial\Delta), \tag{3.20}$$

with $\mu_g > 0$. There is a constant $c_K > 0$ such that

$$\mu_\psi \ge c_K \quad \text{whenever } \psi(0) \in K. \tag{3.21}$$

For the C^2 case, use the real-analytic exhaustion in [7, §7, pp. 212–213]. For centers in K, Lemma 7.2 gives common geometric bounds: the distance of the centers from the boundary is uniformly positive, and in a strongly convex domain the required connecting paths can be chosen as

segments. Propositions 4.6–4.9 then give a common positive lower bound on the stationary weights and uniform continuity of discs, duals, and weights. Apply this approximation to an extremal with prescribed center and derivative direction. Uniformly convergent subsequences preserve the boundary normal relation and $g' \cdot \widetilde{g} = 1$; uniqueness of the normalized extremal identifies the limit with the specified geodesic. This proves (3.20)–(3.21), including uniformity in the center. The left inverse used below is provided by [7, Proposition 3.1.1]. No boundary derivatives of order two of g are required.

Let $d\sigma$ denote normalized arclength on $\partial\Delta$. For an arbitrary $f : \Delta \to D$, define

$$\eta(f,g) = 1 - \operatorname{Re}\big[f'(0)\cdot\widetilde{g}(0) + (f(0)-g(0))\cdot\widetilde{g}'(0)\big]\,. \tag{3.22}$$

Then

$$\eta(f,g) \geq m_D \int_{\partial\Delta} \mu_g |f-g|^2\, d\sigma \geq 0. \tag{3.23}$$

Indeed, apply (3.19) to radial boundary values and integrate. For the bounded holomorphic function

$$A(\zeta) = (g(\zeta) - f(\zeta))\cdot\widetilde{g}(\zeta),$$

the integral of $\operatorname{Re}(A(\zeta)/\zeta)$ is $\operatorname{Re} A'(0) = \eta(f,g)$, by extraction of the first Fourier coefficient. This argument permits different centers; it does not require A/ζ to be holomorphic.

Suppose $g(\Delta)\cap K \neq \varnothing$. Write

$$g = \psi\circ T_\alpha, \qquad T_\alpha(\zeta) = \frac{\alpha+\zeta}{1+\overline{\alpha}\zeta}, \qquad \psi(0)\in K.$$

The normalized dual transforms as

$$\widetilde{g} = \frac{\widetilde{\psi}\circ T_\alpha}{T_\alpha'}, \qquad \mu_g(\zeta) = \mu_\psi(T_\alpha\zeta)\frac{|1+\overline{\alpha}\zeta|^2}{1-|\alpha|^2}.$$

If $d_K = \operatorname{dist}(K,\partial D)$, (3.16) implies

$$1-|\alpha|^2 \leq \frac{4\delta_D(g(0))}{d_K}.$$

Consequently (3.23) bounds the H^2 norm of

$$\frac{(1+\overline{\alpha}\zeta)(f-g)(\zeta)}{\sqrt{1-|\alpha|^2}},$$

and the H^2 point-evaluation estimate gives

$$\sup_{|\zeta|\leq r}|f(\zeta)-g(\zeta)| \leq \frac{2\sqrt{\delta_D(g(0))\,\eta(f,g)}}{\sqrt{m_D c_K d_K}\,(1-r)\sqrt{1-r^2}}. \tag{3.24}$$

In particular, the constant for $r = 1/2$ is

$$\frac{8}{\sqrt{3 m_D c_K d_K}}.$$

All subsequent derivative estimates follow from interior Cauchy estimates; no additional boundary regularity is used.

We now fix the compact sets needed for the targets. Apply (3.1)–(3.9) to each normalized D_i, obtaining constants V_i, and fix $o_i \in D_i$. Put

$$\sigma_i = \min\left\{\frac{\delta_{D_i}(o_i)}{2}, \left(\frac{r_i}{V_i+1}\right)^2\right\}, \qquad K_i = \{x \in D_i : \delta_{D_i}(x) \geq \sigma_i\}. \tag{3.25}$$

A complex geodesic starting in $B(\xi_i, 2r_i)$ and leaving $B(\xi_i, 4r_i)$ meets K_i. Otherwise the real geodesic joining its initial point to an escaping point would, by (3.9), have Euclidean length at most $V_i\sqrt{\sigma_i} < r_i$, although its endpoints are at least $2r_i$ apart.

Choose support constants m_i and weight constants c_{K_i} as in (3.19)–(3.21), and set

$$K_{\mathrm{rig}} = \max_i \frac{8}{\sqrt{3m_i c_{K_i}\sigma_i}}. \tag{3.26}$$

3.4. **The local estimate with a vanishing boundary coefficient.** We prove that there are fixed $c_{\mathrm{loc}} > 0$ and $C_{\mathrm{loc}} \geq 1$ such that

$$\begin{aligned} &0 < h = k_\Omega(z,w) \leq c_{\mathrm{loc}}\delta(z) \\ \implies\quad &\text{a chain of at most } 2n+3 \text{ discs has cost} \\ &< h + C_{\mathrm{loc}}\sqrt{\delta(z)}\,h^2. \end{aligned} \tag{3.27}$$

First consider $a = \delta(z) \leq a_0$. Define

$$\begin{gathered} G = 8M^3, \qquad R = \sqrt{1+14M^2}, \\ \mathcal{A} = \max\{1, 512MK_{\mathrm{rig}}\sqrt{2M}\}, \qquad \mathcal{B} = \sqrt{(1+16M^3)(G+1)}, \\ \Lambda = 16E^2, \qquad U = 2\mathcal{A}(\mathcal{B}+R), \end{gathered}$$

$$c_{\mathrm{run}} = \min\left\{\frac{1}{16M^2}, \frac{1}{4\Lambda U}, \frac{1}{\sqrt{8\mathcal{A}\mathcal{B}}}\right\}, \qquad C_* = 8\mathcal{A}\mathcal{B}(\Lambda+1). \tag{3.28}$$

Assume $0 < h \leq c_{\mathrm{run}}a/2$.

Use the chart selected in §3.2, omitting its index, and put

$$H_D = k_D(\Phi z, \Phi w) \leq h.$$

Let g be a complex geodesic with

$$g(0) = \Phi z, \qquad g(\tanh H_D) = \Phi w.$$

If $g(\Delta) \subset B(\xi, 4r)$, then $\Psi \circ g$ is a single disc in Ω. Its cost is H_D, so necessarily $H_D = h$, proving (3.27) in this case.

Otherwise g meets the fixed compact set K from (3.25), and (3.24) applies.

The target-distance gap satisfies

$$0 \leq h - H_D \leq Ga^2h. \tag{3.29}$$

To see this with the correct direction of comparison, use (3.14) along

$$y(s) = g(\tanh s), \qquad 0 \leq s \leq H_D.$$

This segment lies in $B(\xi, 2r)$, by the independent containment estimates of §3.2, and (3.16)–(3.17) give $\delta_D(y(s)) \le 2Ma$. Since the local target domain lifts into Ω,

$$h \le k_{D\cap B(\xi,4r)}(\Phi z, \Phi w) \le (1 + 4M^3a^2)H_D.$$

This proves (3.29).

Choose an actual finite chain from z to w, of cost S, with

$$h \le S < h + \min\{h, a^2h, \sqrt{a}\,h^2\}. \tag{3.30}$$

Thus $S < c_{\mathrm{run}}a$. The containment estimates of §3.2 also apply to the reference geodesic up to time S, including the short portion beyond H_D. Parametrize the chain's disc arcs by source Poincaré arclength:

$$\gamma : [0, S] \to \Omega.$$

At each nonjunction time t, recenter the current original disc as f_t, so that

$$f_t(0) = \gamma(t), \qquad f_t'(0) = \gamma'(t).$$

The family is continuous on each of the finitely many closed chain intervals; retain both choices at junctions when taking its compact range.

Set

$$F_t = \Phi \circ f_t, \qquad g_t = g \circ T_{\tanh t}, \qquad q_t = \sqrt{\eta(F_t, g_t)}.$$

The key estimate is

$$\int_0^S q_t\,dt \le \mathcal{B}aS, \qquad q_t \le \frac{R}{\sqrt{a}}. \tag{3.31}$$

Here are the details of the moving-center calculation. Write

$$X(t) = \Phi(\gamma(t)), \qquad L(t) = \frac{\widetilde{g}(\tanh t)}{1 - \tanh^2 t}.$$

The automorphism transformation law gives

$$L(t) = \widetilde{g}_t(0), \qquad L'(t) = \widetilde{g}_t'(0), \qquad y'(t) \cdot L(t) = 1.$$

Indeed, with $u = \tanh t$, direct differentiation gives

$$L'(t) = \widetilde{g}'(u) + \frac{2u}{1-u^2}\widetilde{g}(u) = \widetilde{g}_t'(0).$$

Therefore, on every chain interval,

$$\eta(F_t, g_t) = -\frac{d}{dt}\,\mathrm{Re}\big[(X(t) - y(t)) \cdot L(t)\big]. \tag{3.32}$$

The expression being differentiated is continuous at junctions. Thus, with $\Delta = S - H_D$,

$$\int_0^S \eta(F_t, g_t)\,dt = \mathrm{Re}\big[(y(S) - y(H_D)) \cdot L(S)\big]. \tag{3.33}$$

The normalized Lempert left inverse π_g satisfies $\pi_g \circ g = \mathrm{id}_\Delta$ and $d\pi_g(g(\zeta)) = \widetilde{g}(\zeta)$. Schwarz–Pick on the Euclidean ball centered at $g(\zeta)$, followed by (3.16), gives

$$|\widetilde{g}(\zeta)| \le \frac{1-|\zeta|^2}{\delta_D(g(\zeta))} \le \frac{(1+|\zeta|)^2}{\delta_D(g(0))}.$$

Cauchy's estimate on circles of radius tending to one therefore gives

$$|\widetilde{g}(0)| \le \delta_D(g(0))^{-1}, \qquad |\widetilde{g}'(0)| \le 4\delta_D(g(0))^{-1}.$$

The same estimates apply to g_t. Its center has depth between $a/(2M)$ and $2Ma$, so (3.22), the diameter bounds, and Cauchy's estimate imply

$$q_t^2 \le \frac{1+10M^2}{a} \le \frac{R^2}{a}.$$

For the integrated estimate, (3.15) and Cauchy on circles of radius $3/8$, centered at $\tanh s$, give

$$|y''(s)| \le \frac{131}{9} M^{3/2}\sqrt{a}, \qquad 0 \le s \le S \le 1/16.$$

Also $|L(S)| \le 2M/a$. Taylor expansion at S in (3.33) therefore gives

$$\left| \int_0^S \eta(F_t, g_t)\, dt - \Delta \right| \le 16M^3 \frac{\Delta^2}{\sqrt{a}}.$$

By (3.29)–(3.30),

$$0 \le \Delta \le (G+1)a^2 S, \qquad \frac{\Delta}{\sqrt{a}} \le 1.$$

For the second inequality, note explicitly that

$$\frac{\Delta}{\sqrt{a}} \le \frac{9M}{16} a^{5/2} \le 1, \qquad a \le a_0 \le \frac{1}{4096M^4},$$

using $G+1 \le 9M^3$ and $S \le a/(16M^2)$. Cauchy–Schwarz now proves (3.31).

We next pull the estimate back through the inverse chart. All half-discs involved are already in the fixed chart, as established in §3.2. Put

$$\mathcal{G}(s) = \Psi(g(\tanh s)), \qquad H_t(s) = f_t(\tanh(s-t)), \qquad e_t(s) = H_t(s) - \mathcal{G}(s).$$

For $0 \le s, t \le S$, estimate (3.24), the inverse derivative bound, and interior Cauchy estimates give

$$|e_t'(s)|,\ |e_t''(s)| \le \mathcal{A}\sqrt{a}\, q_t. \tag{3.34}$$

To verify the numerical constant, on $|\zeta| \le 1/4$ consider

$$f_t(T_{-\tanh t}\zeta) - \Psi(g(\zeta)).$$

Its norm is at most

$$MK_{\mathrm{rig}}\sqrt{2Ma}\, q_t.$$

Cauchy on circles of radius $1/8$, followed by differentiation of $\zeta = \tanh s$, bounds its first and second s-derivatives by 8 and 129 times that quantity. The choice of $\mathcal{A}$ in (3.28) is therefore sufficient.

Since $\gamma'(t) - \mathcal{G}'(t) = e_t'(t)$, (3.31)–(3.34) imply

$$|\gamma(t) - \mathcal{G}(t)| \le \mathcal{A}\mathcal{B}a^{3/2}S. \tag{3.35}$$

Apply Carathéodory's finite-dimensional convex-hull theorem to the compact range of

$$t \longmapsto (e_t'(0), q_t) \in \mathbb{R}^{2n+1}.$$

It supplies at most $N \le 2n+2$ times t_j and positive weights ℓ_j such that

$$\sum_j \ell_j = S, \qquad \sum_j \ell_j e_{t_j}'(0) = \int_0^S e_t'(0)\,dt, \qquad \sum_j \ell_j q_{t_j} = \int_0^S q_t\,dt. \tag{3.36}$$

This is an exact finite-dimensional convex-hull statement about one finite chain; no extremal-chain attainment or limiting selection is involved.

Let $s_j = \sum_{i\le j} \ell_i$, and define

$$x_0 = z, \qquad x_j = x_{j-1} + H_{t_j}(s_j) - H_{t_j}(s_{j-1}).$$

Since $H_t' = \mathcal{G}' + e_t'$, the reference increments telescope, and (3.34)–(3.36) give

$$|x_j - \mathcal{G}(s_j)| \le \mathcal{A}\mathcal{B}a^{3/2}S.$$

At the final endpoint, preservation of the first moment in (3.36) gives the stronger bound

$$|x_N - w| \le 2\mathcal{A}\mathcal{B}a^{3/2}S^2. \tag{3.37}$$

Indeed, subtract the common moment from

$$x_N - \mathcal{G}(S) = \sum_j \int_{s_{j-1}}^{s_j} e_{t_j}'(s)\,ds, \qquad w - \mathcal{G}(S) = \int_0^S e_t'(t)\,dt,$$

and use the second derivative bound in (3.34). Each remainder is at most

$$\mathcal{A}\sqrt{a}\,S \int_0^S q_t\,dt \le \mathcal{A}\mathcal{B}a^{3/2}S^2.$$

Thus the extra factor S in (3.37) comes from exact preservation of the derivative moment, rather than from a pointwise closeness assertion.

Define the translation required for link j by

$$v_j = x_{j-1} - H_{t_j}(s_{j-1}).$$

Equations (3.31), (3.34), and (3.35) give

$$|v_j| \le 2\mathcal{A}\mathcal{B}a^{3/2}S + 2\mathcal{A}\sqrt{a}\,Sq_{t_j} \le US,$$

and, crucially,

$$\sum_j \ell_j |v_j| \le 4\mathcal{A}\mathcal{B}a^{3/2}S^2. \tag{3.38}$$

Recenter f_{t_j} at source time $s_{j-1} - t_j$, obtaining a whole disc D_j. Its center has depth at least $a/(4E)$, by (3.3), because the original chain centers have

depth at least $a - S$, and the recentering distance is at most S. Thus (3.4) gives

$$\operatorname{dist}(D_j(r\Delta), \partial\Omega) \geq \frac{a(1-r)}{8E^2}.$$

Set

$$r_j = 1 - \frac{\Lambda|v_j|}{a}.$$

If $v_j = 0$, leave the original whole disc unchanged. Otherwise our choice of c_{run} ensures $3/4 \leq r_j < 1$, and the preceding clearance is at least $2|v_j|$. Hence in either case

$$\zeta \longmapsto D_j(r_j\zeta) + v_j$$

is a holomorphic disc entirely contained in Ω.

Its endpoint parameter is

$$\frac{\tanh \ell_j}{r_j}.$$

This preserves the prescribed increment exactly. For $\ell_j \leq 1/8$ and $r_j \geq 3/4$,

$$\operatorname{arctanh}\left(\frac{\tanh \ell_j}{r_j}\right) - \ell_j \leq 2\ell_j(1 - r_j).$$

Using (3.38), the total additional cost is at most

$$8\Lambda\mathcal{A}\mathcal{B}\sqrt{a}\, S^2.$$

Finally, (3.37) is corrected by one affine disc in $B(w, a/2)$. The smallness condition in (3.28) makes its endpoint parameter at most $1/2$, and its cost is at most

$$8\mathcal{A}\mathcal{B}\sqrt{a}\, S^2.$$

The assembled chain therefore uses at most $2n + 3$ discs and has cost

$$\leq S + C_*\sqrt{a}\, S^2 < h + (1 + 4C_*)\sqrt{a}\, h^2. \tag{3.39}$$

3.5. **Extension of the local estimate to the compact interior.** We give a separate elementary estimate. For any $a = \delta(z)$, if

$$0 < h = k_\Omega(z, w) \leq \frac{a}{128E},$$

there is a chain of at most $2n + 2$ discs with cost

$$< h + \frac{128E}{a}h^2. \tag{3.40}$$

Choose a finite chain with

$$h \leq S < h + \min\{h/16, Eh^2/(16a)\},$$

and use centered discs f_t as above. Put

$$r_1 = 1 - \frac{8ES}{a}, \qquad r_2 = 1 - \frac{16ES}{a}.$$

Since $|\gamma(t) - z| \leq S$, equation (3.4) shows that

$$F_t(\zeta) = f_t(r_1\zeta) + z - \gamma(t)$$

maps Δ into Ω and is centered at z: before translation the contracted disc has clearance at least $2S$.

Apply the finite-dimensional convex-hull theorem in $\mathbb{R}^{2n}$ to $f_t'(0)$, whose integral is $w-z$, and divide the resulting weights by r_1. We obtain at most $2n+1$ selected discs and weights $b_j \geq 0$ satisfying

$$\sum_j b_j F_j'(0) = w - z, \qquad s := \sum_j b_j = S/r_1.$$

Concatenate translated copies of $F_j(r_2\zeta)$, using endpoint parameters b_j/r_2.

Here

$$\frac{ES}{a} \leq \frac{1}{64}, \qquad r_1 \geq \frac{7}{8}, \quad r_2 \geq \frac{3}{4}, \quad s \leq \frac{1}{56}.$$

The Taylor coefficients of $F_j - z$ have norm at most 1. Hence the cumulative translation is bounded by

$$\frac{s}{1-s} < 2S,$$

whereas the second contraction has clearance at least $8S$. The discs therefore remain in Ω. The final endpoint error is less than

$$\frac{s^2}{1-s} < 3S^2.$$

Furthermore,

$$\frac{1}{r_1 r_2} \leq 1 + 34ES/a, \qquad \sum_j \left(\frac{b_j}{r_2}\right)^3 < 4S^3.$$

Using $\operatorname{arctanh} u \leq u + u^3$ for the present parameters, the link cost is at most

$$S + 38ES^2/a.$$

An affine correction in $B(w, a/2)$ costs at most $12S^2/a$. The total is therefore at most

$$S + 50ES^2/a < h + 128Eh^2/a,$$

proving (3.40).

Combining (3.39) and (3.40), fix

(3.41)

$$\boxed{c_{\mathrm{loc}} = \min\left\{\frac{c_{\mathrm{run}}}{2}, \frac{1}{128E}\right\}, \qquad C_{\mathrm{loc}} = \max\left\{1 + 32\mathcal{A}\mathcal{B}(\Lambda+1), \frac{128E}{a_0^{3/2}}\right\}.}$$

Then (3.27) holds for every distinct pair $z, w \in \Omega$. The global approximation theorem for $z = w$ is immediate from a constant disc.

3.6. **Boundary caps.** Choose

$$\rho = \frac{1}{4} \min_j \operatorname{dist}(\overline{W_j}, \mathbb{C}^n \setminus V_j) > 0$$

for the cover used in (3.11), and set

$$d_0 = \min\left\{1, \frac{\delta(o)}{2}, \left(\frac{\rho}{V+1}\right)^2\right\}, \qquad J = \frac{LW^2}{2}. \tag{3.42}$$

A real geodesic segment of maximum depth at most $d \le d_0$ lies in a single V_j. Indeed, (3.9) places the whole segment within

$$d + V\sqrt{d} \le \rho$$

of a nearest boundary point to one endpoint.

Along any unit-speed real Kobayashi geodesic,

$$\gamma_\Omega(\gamma(s); \gamma'(s)) \le 1 \quad \text{almost everywhere.}$$

At each point where γ is Euclidean differentiable, this follows by differentiating the contraction inequality for every holomorphic map $\Omega \to \Delta$, then taking the supremum. The same differentiability set works for all such maps.

Integrating (3.11), using (3.9), and applying Lempert's equality on the convexifiable domain $\Omega \cap U_j$, we obtain

$$\ell_\Omega(x, y) \le k_\Omega(x, y) + Jd^2. \tag{3.43}$$

Here ℓ_Ω is the one-disc Lempert function. Thus, for every $\eta > 0$, one actual disc joins the cap endpoints with cost less than

$$k_\Omega(x, y) + Jd^2 + \eta.$$

Lempert's equality for convex domains requires no smoothness of the convex intersection; see [11, Theorem 1].

3.7. **Weighted subdivision and the final constant.** Define

$$d_* = \min\left\{d_0, [8(J+1)]^{-1/2}\right\}, \qquad m_* = bd_*,$$

and

$$\lambda = \min\left\{\frac{2c}{3} m_*^{3/4}, \quad c_{\mathrm{loc}}\, 2^{-1/3} m_*^{5/4}, \quad \frac{1}{4\, 2^{2/3} C_{\mathrm{loc}} Q_0}\right\} > 0. \tag{3.44}$$

The constant in the theorem is

$$\boxed{A_\Omega = \frac{Q_0}{\lambda}.} \tag{3.45}$$

Fix $0 < \varepsilon \le 1$, and put

$$d = \min\left\{d_0, \sqrt{\frac{\varepsilon}{8(J+1)}}\right\}.$$

If $z = w$, a constant disc proves the theorem. Assume henceforth that $z \ne w$, and let γ be a real geodesic from z to w.

If its maximum depth is at most d, equation (3.43) gives a single disc of the required cost. Otherwise split it at its first and last visits to

$$\{\delta \geq d\}.$$

The initial and terminal pieces are shallow caps; either may be empty.

Every point x of the middle segment satisfies

$$\delta(x) \geq bd \geq m_*\sqrt{\varepsilon}. \tag{3.46}$$

Indeed, an H-thin triangle with third vertex o gives

$$k_\Omega(o,x) \leq \max\{k_\Omega(o,x_-), k_\Omega(o,x_+)\} + H$$

for its middle endpoints x_-, x_+. Their depths are at least d, so (3.1) gives (3.46). This argument allows arbitrary repeated boundary excursions between the first and last visits.

On the middle segment introduce

$$u(s) = \int_0^s \delta(\gamma(t))^{1/4}\, dt.$$

By (3.6) and the Lipschitz property of δ,

$$\left|\frac{d}{du}\delta^{3/4}\right| \leq \frac{3}{4c} \quad \text{almost everywhere.} \tag{3.47}$$

Its total u-length is at most Q_0, by (3.10). Partition it into intervals of u-length

$$\tau_i \leq \tau := \lambda\varepsilon.$$

The number of intervals is at most

$$\left\lceil \frac{Q_0}{\lambda\varepsilon} \right\rceil.$$

Let a_i be the initial depth of one interval, and h_i its Kobayashi length. Equations (3.44), (3.46), and (3.47) give

$$\delta \geq 2^{-4/3} a_i \quad \text{on that interval,}$$

and therefore

$$h_i \leq 2^{1/3}\tau_i a_i^{-1/4} \leq c_{\mathrm{loc}} a_i.$$

For the last inequality, use $a_i \geq m_*\sqrt{\varepsilon}$ and $0 < \varepsilon \leq 1$.

Apply (3.27) to its endpoints. The resulting excess is less than

$$C_{\mathrm{loc}}\sqrt{a_i}\, h_i^2 \leq 2^{2/3} C_{\mathrm{loc}} \tau_i^2.$$

Summing,

$$\text{middle excess} \leq 2^{2/3} C_{\mathrm{loc}} \tau Q_0 \leq \frac{\varepsilon}{4}. \tag{3.48}$$

For each nonempty cap, use (3.43) with disc-infimum slack $\varepsilon/16$. Their combined excess is less than

$$2Jd^2 + \frac{\varepsilon}{8} \leq \frac{3\varepsilon}{8}. \tag{3.49}$$

The complete chain consequently has cost less than

$$k_\Omega(z,w) + \frac{5\varepsilon}{8} < k_\Omega(z,w) + \varepsilon.$$

It uses at most two cap discs and $2n+3$ discs per middle interval. Equations (3.45) and (3.48) give exactly (1.5). □

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